

Trainings ▾

Preparatory Steps

Rear Gear Train

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Disconnect the batteries. Refer to Procedure 013-009 in Section 013. (</qs3/pubsys2/xml/en/procedures/40/40-013-009.html>)
- Remove the engine water pump (cooling fan) drive belt. Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html>)
- Remove the vibration damper/crankshaft pulley.
- If equipped, remove the viscous damper. Refer to Procedure 001-052 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html>)
- If equipped, remove the rubber damper. Refer to Procedure 001-051 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html>)
- If equipped, remove the crankshaft pulley **only**. Refer to Procedure 001-022 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html>)
- If required, remove the fan hub pulley. Refer to Procedure 008-039 in Section 8. (</qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html>)
- Remove the front crankshaft seal. Refer to Procedure 001-023 in Section 1.

Front Gear Train

WARNING

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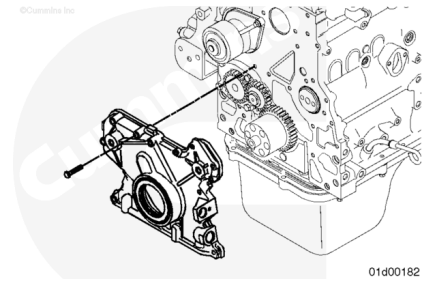
- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/40/40-013-009.html)
- Remove the engine water pump (cooling fan) drive belt. Refer to Procedure 008-002 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)
- If required, remove the fan hub pulley. Refer to Procedure 008-039 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html)
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- If equipped, remove the rubber damper. Refer to Procedure 001-051 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html)
- If equipped with a crankshaft pulley **only**. Refer to Procedure 001-022 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html)

Remove

Rear Gear Train

⚠ CAUTION ⚠

Some rear gear train engines, with a cast aluminum front gear cover, have two threaded holes to aid in the removal of the front gear cover. When looking at the front of the front gear cover, there is a threaded through hole in the lower right-hand corner that can be used with a jackscrew. On the lower left-hand corner, there is a threaded blind hole. A capscrew can be inserted into that blind hole to be used with a slide hammer. Do NOT use the lower left-hand threaded blind hole as a jackscrew or the front gear cover will be damaged.



Remove the front gear cover mounting capscrews.

⚠ CAUTION ⚠

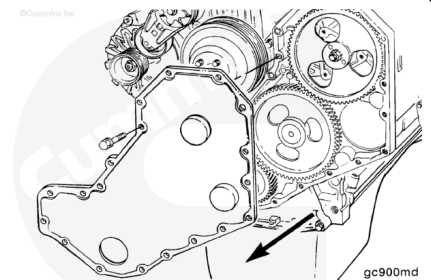
To break the seal, pry the front gear cover away from the front gear housing . Be careful NOT to damage the front gear cover when breaking the seal to the front gear housing.

Remove the front gear cover.

Front Gear Train

⚠ CAUTION ⚠

To break the seal, pry the front gear cover away from the front gear housing . Be careful NOT to damage the front gear cover when breaking the seal to the front gear housing.

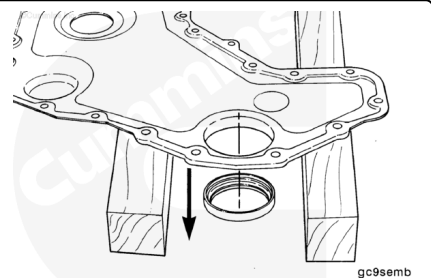


Remove the front gear cover mounting capscrews.

Remove the front gear cover with the front crankshaft seal.

Note : Some engines have an additional dust shield installed in front of the front crankshaft seal.

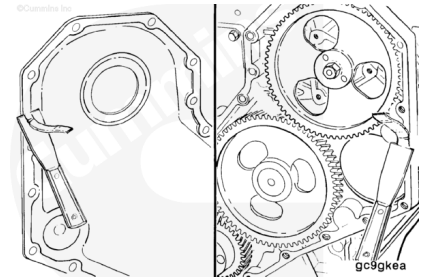
Remove the front crankshaft seal. Refer to Procedure 001-023 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-023-tr.html>)



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Scrape the sealant from the front gear cover and gear housing surface for front gear train engines, or the cylinder block, for rear gear train engines.

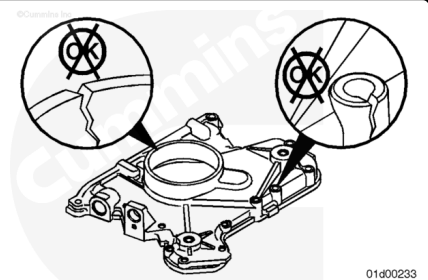
Use solvent or steam to clean the front gear cover.

Dry the cover with compressed air.

Inspect the gear cover for cracks or other damage.

Replace the front gear cover if **any** damage is found.

Note : Parts from the damaged front gear cover can be used again when replacing the front gear cover. Examples are the oil fill neck, fuel pump gear access plate, etc.

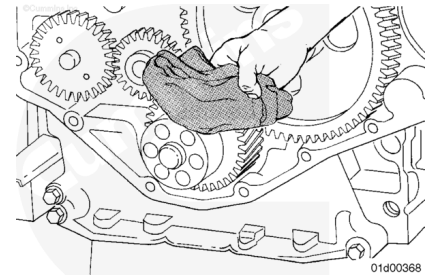


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When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Use solvent to clean the oil and seal residue from the crankshaft surface.

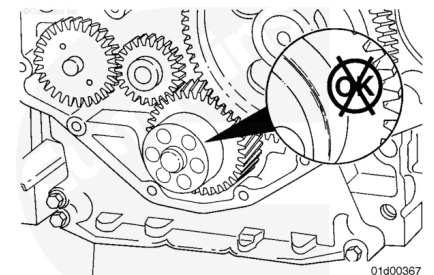
Dry with compressed air.

Note : If the crankshaft has excessive wear, a service wear sleeve is available for engines that use a lip style front crankshaft seal.

Inspect the nose of the crankshaft for excessive wear.

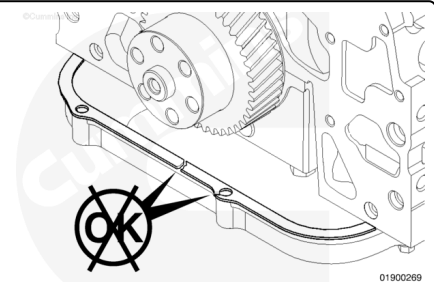
For engines equipped with a non-lip style front crankshaft seal, use a fine crocus cloth to remove any nicks or burrs. There is **no** wear sleeve available if the crankshaft nose sealing surface is damaged.

For engines equipped with a lip style front crankshaft seal, inspect the crankshaft seal contact area for a wear groove. If the groove is deeper than 0.25 mm [0.010 in], a wear sleeve and oversize seal **must** be used. Refer to Procedure 001-025 in Section 1. (</qs3/pubsys2/xml/en/procedures/40/40-001-025-tr.html>)



For rear gear train engines, inspect the oil pan gasket for damage.

If the engine is equipped with a suspended oil pan with a rubber oil pan gasket, and the gasket is found to be damaged, the oil pan **must** be removed and the entire oil pan gasket replaced. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/40/40-007-025-tr.html>)

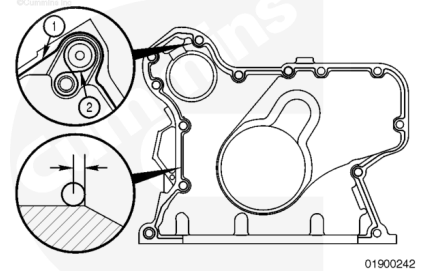


Install

Rear Gear Train

Note : Install three guide pins, Part Number 3164977, to improve alignment of the front cover and front seal to the gear housing and crankshaft.

Apply a 1.5 to 2.0 mm [0.06 to 0.08 in] wide bead of sealant, Part Number 3164070, to the block side of the front gear cover in the path illustrated.



On engines equipped with an oil pressure sensor/switch located in the front gear cover, there are two critical sealant paths that **must** be followed:

1. In this area, the sealant **must** be applied towards the outer edge of the front gear cover to avoid an unused mounting hole in the cylinder block.

Note : To make sure the unused mounting hole in the cylinder block does not affect the sealing joint, fill the mounting hole with sealant, Part Number 3164070.

2. Sealant **must** be applied around both the mounting hole location and the oil supply hole in the front gear cover.

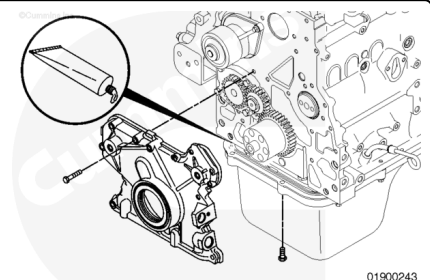
Note : Install the front cover within 10 minutes of applying the sealant, or the sealant will not seal correctly. Once installed, allow the sealant to dry for 30 minutes before operating the engine.

Note : Before installing the front gear cover, make sure sealant, Part Number 3164067, has been applied to the intersecting joint of the cylinder block, oil pan, and front gear cover.

Use the dowel rings to locate the front gear cover and install the front gear cover onto the cylinder block.

Install the front gear cover to the cylinder block using mounting capscrews and the oil pan to the front gear cover using mounting capscrews. Do **not** tighten the capscrews at this time.

If installed, remove the thread or wire holding the oil pan paper gasket in place.

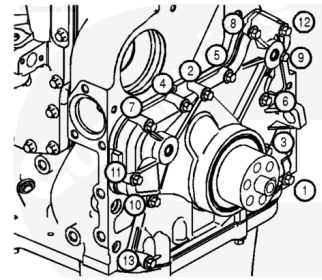


Tighten the front gear cover to the cylinder block mounting capscrews in the order indicated.

Tighten the oil pan-to-front gear cover mounting capscrews, starting with the inner capscrews first.

Note : Depending on the type of lubricating oil pan, there are two or four mounting capscrews.

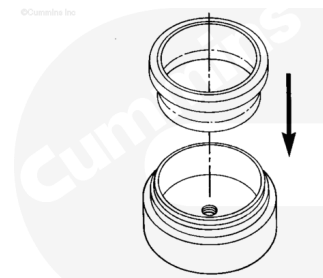
Torque Value: 24 n•m [212 in-lb]



Front Gear Train

Install a new front crankshaft seal into the front gear cover prior to installing the front gear cover. Refer to Procedure 001-023 in Section 1. (/qs3/pubsys2/xml/en/procedures/40/40-001-023-tr.html)

Leave the plastic pilot installation tool in the front crankshaft seal.



⚠ CAUTION ⚠

The seal lip and the sealing surface on the crankshaft must be free from all oil residue to prevent seal leaks.

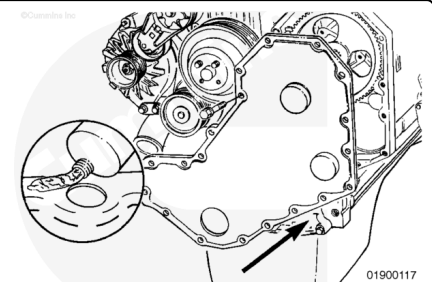
Stamped Steel Gear Cover - Apply a thin bead of sealant, Part Number 3164070, to the sealing surface of the front cover.

Note : Do not remove the plastic seal pilot tool from the front crankshaft seal at this time. Use the plastic seal pilot tool to guide the seal onto the crankshaft.

Note : Install the front cover within 10 minutes of applying the sealant, or the sealant will not seal correctly. Once installed, allow the sealant to dry for 30 minutes before operating the engine.

Cast Aluminum Gear Cover - Place the gasket, Part Number 3918673, between the gear cover and the gear housing.

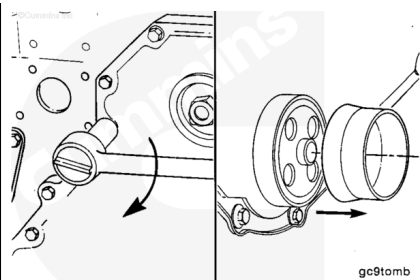
Install the front gear cover on the engine, with the plastic seal pilot tool to guide the front crankshaft seal onto the crankshaft.



Install and tighten the front gear cover capscrews.

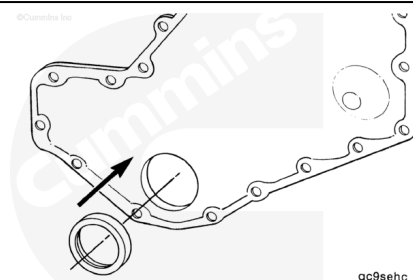
Torque Value: 24 n•m [212 in-lb]

Remove the plastic pilot tool from the crankshaft.



If previously equipped, or if the engine operates in a dusty environment, install a dust shield.

A pilot tool is **not** necessary for the dust shield. Slide the dust shield over the nose of the crankshaft.



Finishing Steps

Rear Gear Train

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Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- If the oil pressure sensor/switch is located in the front gear cover, install and connect the sensor/switch. Refer to the OEM service manual.
- Install the crankshaft front seal. Refer to Procedure 001-023 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-023-tr.html)
- If required, install the fan hub pulley. Refer to Procedure 008-039 in Section 8.
(/qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html)
- Install the vibration damper/crankshaft pulley. Refer to Procedure 001-052 in Section 1.

(/qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html)

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(/qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html)
- If equipped with a crankshaft pulley **only** use the following procedure. Refer to Procedure 001-022 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/40/40-013-009.html)
- Operate engine and check for leaks.

Front Gear Train

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- Install the vibration damper/crankshaft pulley. Refer to Procedure 001-052 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-052-tr.html)
- If equipped with a rubber damper. Refer to Procedure 001-051 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-051-tr.html)
- If equipped with a crankshaft pulley **only**. Refer to Procedure 001-022 in Section 1.
(/qs3/pubsys2/xml/en/procedures/40/40-001-022-tr.html)
- If removed, install the fan hub. Refer to Procedure 008-039 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-039-tr.html)
- Install the engine water pump (cooling fan) drive belt. Refer to Procedure 008-002 in Section 8.
(/qs3/pubsys2/xml/en/procedures/40/40-008-002-tr.html)

- Connect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/40/40-013-009.html>)
- Operate the engine and check for leaks.

Last Modified: 03-Aug-2016
